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Domain :- AI - ML

Branch Name :- Computer Engineering

Insurance Dataset

1) Load the insurance dataset using pandas and display the first 10 rows of the data.

[20]

```
df.head(10)
```

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86
5	31	female	25.7	0	no	southeast	3756.62
6	46	female	33.4	1	no	southeast	8240.59
7	37	female	27.7	3	no	northwest	7281.51
8	37	male	29.8	2	no	northeast	6406.41
9	60	female	25.8	0	no	northwest	28923.14

Next steps: Generate code with df New interactive sheet

2) Use pandas functions to get basic information about the dataset:
   Shape of the data (shape)
   Column names and data types (info())
   Statistical summary of numerical columns (describe().

a) Shape

```
[22]
0s
df.shape
```

```
(1338, 7)
```

b) Column names and data types (info())

```
[23]
0s
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   expenses  1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

c) Columns Describe

```
df.describe()
```

	age	bmi	children	expenses
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.665471	1.094918	13270.422414
std	14.049960	6.098382	1.205493	12110.011240
min	18.000000	16.000000	0.000000	1121.870000
25%	27.000000	26.300000	0.000000	4740.287500
50%	39.000000	30.400000	1.000000	9382.030000
75%	51.000000	34.700000	2.000000	16639.915000
max	64.000000	53.100000	5.000000	63770.430000

3) Check if the dataset contains any null/missing values. Show the count of null values for each column.

[26] 0s

```
df.isnull().sum()
```

	0
age	0
sex	0
bmi	0
children	0
smoker	0
region	0
expenses	0

dtype: int64

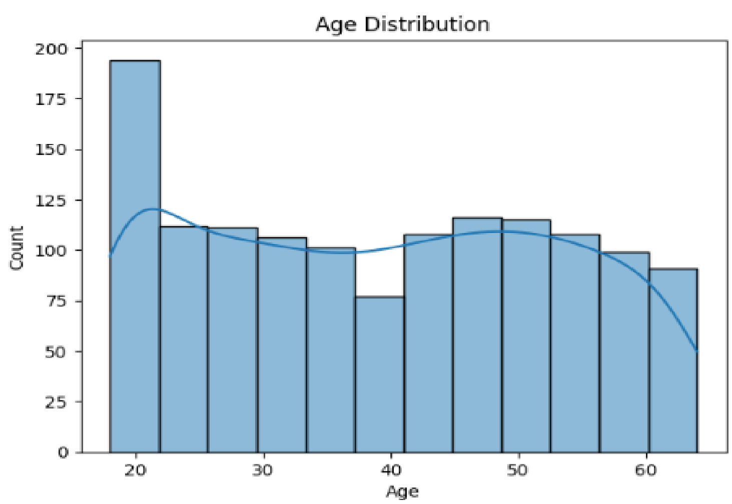
4) Create distribution plots for all numerical columns (age, bmi, children,charges):

   Use sns.histplot() or sns.distplot()

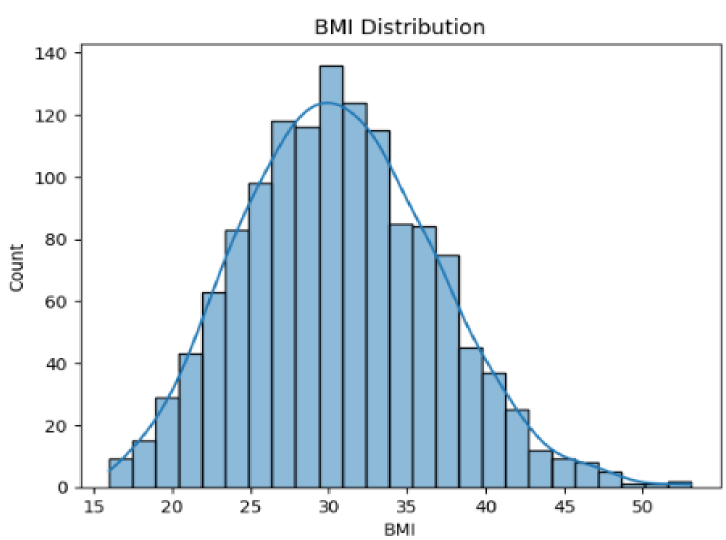
   Create one plot per column (total 4 plots)

   Add meaningful titles and labels

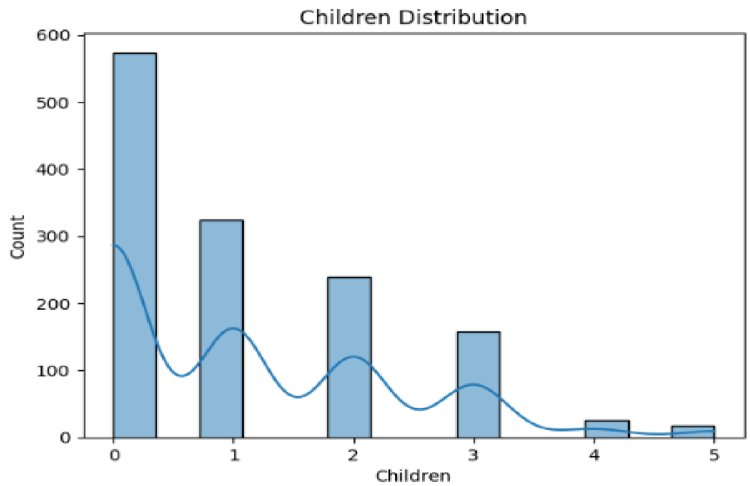
```
sns.histplot(df["age"], kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```




```
sns.histplot(df["bmi"], kde=True)
plt.title("BMI Distribution")
plt.xlabel("BMI")
plt.ylabel("Count")
plt.show()
```



```python
sns.histplot(df["children"], kde=True)
plt.title("Children Distribution")
plt.xlabel("Children")
plt.ylabel("Count")
plt.show()
```

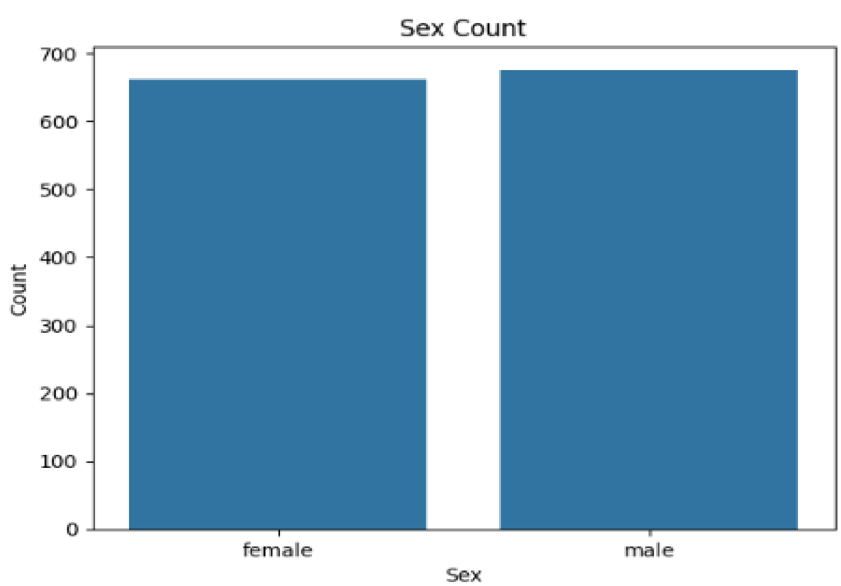


5) Create count plots for all categorical columns (sex, smoker, region):

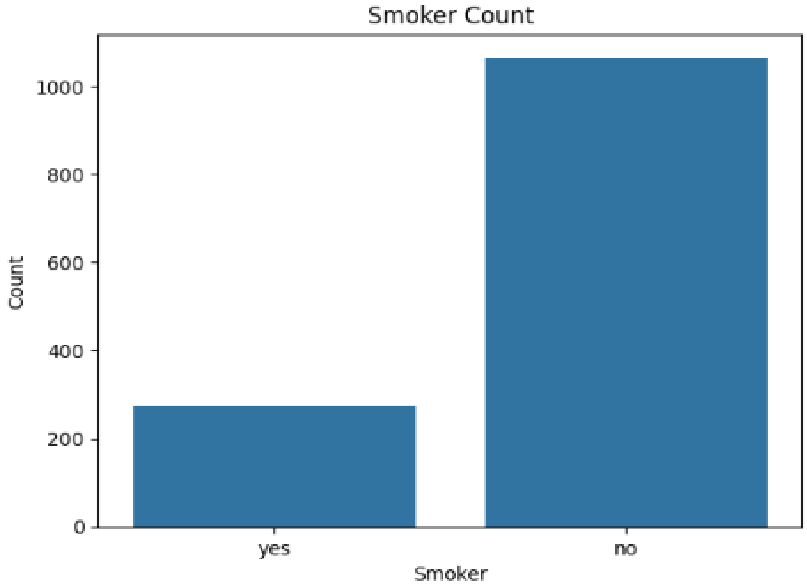
   Use sns.countplot() for each column

   Add titles and labels for clarity

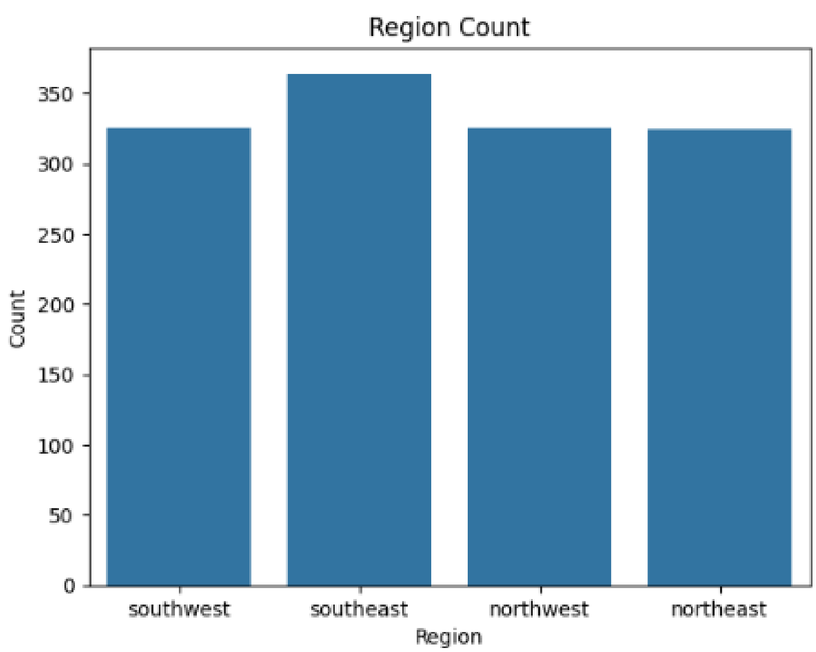
```python
sns.countplot(x=df["sex"])
plt.title("Sex Count")
plt.xlabel("Sex")
plt.ylabel("Count")
plt.show()
```



```
sns.countplot(x=df["smoker"])
plt.title("Smoker Count")
plt.xlabel("Smoker")
plt.ylabel("Count")
plt.show()
```

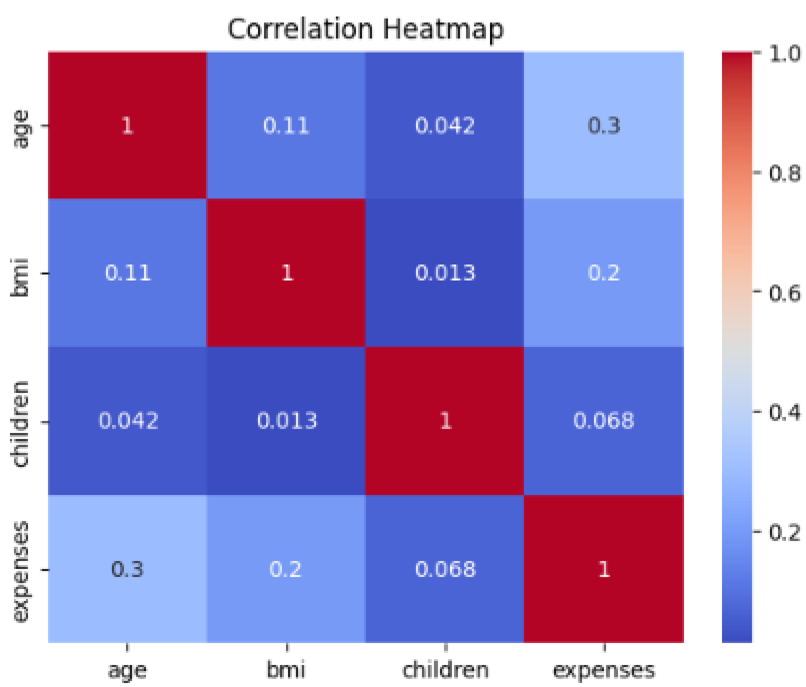



```
sns.countplot(x=df["region"])
plt.title("Region Count")
plt.xlabel("Region")
plt.ylabel("Count")
plt.show()
```



6) Create a heatmap to show the correlation between numerical variables.Use sns.heatmap()
Annotate the values and use a suitable color palette

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



```
sns.histplot(df["age"], bins=15, kde=True, color="green")
plt.title("Age Distribution")
plt.show()
```